

Human Review: Divergent orthoprojections in a strictly convex Banach space

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed construction appears to give a valid answer to the question in Badea-Lyubich asking whether divergent iterates of a product of orthoprojections can occur in a strictly convex Banach space.

The idea is to start from the elementary two-dimensional obstruction and then blow it up to infinite dimensions. For each $0 < \rho < 1$, the proof builds a two-dimensional strictly convex block E_ρ with two norm-one projections P_ρ, Q_ρ such that $P_\rho Q_\rho$ has eigenvalue $-\rho^2$. Taking $\rho_n \uparrow 1$ gives a sequence of blocks whose products behave more and more like multiplication by -1 .

Strict Convexity of the Sum

The key point in the infinite-dimensional construction is the renorming. One starts with the ℓ_∞ -sum

$$Y = \left(\bigoplus_{n=1}^{\infty} E_n \right)_{\ell_\infty},$$

and defines

$$\|x\|_X^2 = \|x\|_\infty^2 + \sum_{n=1}^{\infty} 2^{-n} \|x_n\|_{E_n}^2.$$

This norm is equivalent to the original ℓ_∞ -sum norm, since

$$\|x\|_\infty \leq \|x\|_X \leq \sqrt{2} \|x\|_\infty.$$

The strict convexity comes from the weighted ℓ_2 -term. Since each block E_n is strictly convex, the function

$$x_n \mapsto \|x_n\|_{E_n}^2$$

is strictly convex on E_n . Hence

$$x \mapsto \sum_{n=1}^{\infty} 2^{-n} \|x_n\|_{E_n}^2$$

is strictly convex on Y : if $x \neq y$, then $x_k \neq y_k$ for some coordinate k , and the positive weight 2^{-k} preserves the strict inequality coming from that block. The term $x \mapsto \|x\|_\infty^2$ is convex, so their sum $\|x\|_X^2$ is strictly convex. Therefore the norm $\|\cdot\|_X$ is strictly convex.

This also explains why the construction is not uniformly convex. The weights 2^{-n} tend to zero, so the added curvature becomes arbitrarily weak in far-out coordinates. Thus the renorming gives strict convexity without destroying the asymptotic ℓ_∞ -sum behaviour needed for the counterexample.

Contractivity and Divergence

The blockwise projections

$$P(x_n) = (P_n x_n), \quad Q(x_n) = (Q_n x_n)$$

remain contractive for the new norm. Indeed, they are contractive on each block, hence contractive for both the ℓ_∞ -term and the weighted ℓ_2 -term. Therefore they are norm-one projections for $\|\cdot\|_X$.

The divergence calculation is then straightforward. If u_n is the distinguished eigenvector in the n -th block and $x = (u_n)_n$, then for $T = PQ$,

$$T^m x = ((-\rho_n^2)^m u_n)_n.$$

Since $\rho_n \uparrow 1$, for every fixed m the far-out coordinates make $\rho_n^{4m}(1 + \rho_n^2)$ arbitrarily close to 2. Thus

$$\|T^{2m} x - T^{2m+1} x\|_\infty = 2,$$

and hence

$$\|T^{2m} x - T^{2m+1} x\|_X \geq 2.$$

So the orbit is not Cauchy, and the iterates $(PQ)^m$ do not converge strongly.

Review Outcome

The proof appears correct. The essential mechanism is clear: use two-dimensional almost- ℓ_∞^2 blocks with product eigenvalues $-\rho_n^2$, take $\rho_n \rightarrow 1$, and place the blocks in an ℓ_∞ -sum equipped with a small weighted ℓ_2 -correction. The correction is enough to force strict convexity, but not enough to remove the far-out oscillations responsible for divergence.

One minor point to keep in mind is the scalar field. The proof as written is most direct over the real field. If the intended reading of the source question is complex-linear, the balanced complex variant of the two-dimensional block should be checked explicitly. Apart from this standard-field issue, the argument appears sound.